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Weighted severity Decision Rule (Weighted Severity Index)

Weights

- ① Lesion count $\rightarrow$ most imp for disease burden (highest weight)
- ② Lesion Density $\rightarrow$ local clustering $\rightarrow$ Next imp as it reflects severity in small area
- ③ Redness Index $\rightarrow$ inflammation $\rightarrow$ If few red lesions then these are not severe unless these are inflamed lesions.

Weights

Lesion count + Density + RI
0.5 0.25 0.25

Numeric score

Mild	Moderate	Severe
0	0.5	1

Example

- ① Suppose
count - mild, density - mild, Redness - severe
then
- | | | |
|--------------|--------------|--------------|
| $\downarrow$ | $\downarrow$ | $\downarrow$ |
| 0 | 0 | 1 |

$$\begin{aligned} WSI &= 0.5 \times 0 + 0.25 \times 0 + 0.25 \times 1 \\ &= 0.25 \end{aligned}$$

$$\begin{aligned} \text{WSI} &= 0.5 + 0.25 + 0.25 \\ &= 1 \end{aligned}$$

Further, as we are working with 3 parameters, so we divide 1 by 3 $= \frac{1}{3} = 0.33$

So decision rule

If $\text{WSI} \leq 0.33$ then mild

If $\text{WSI} \leq 0.66$ then moderate

& If $\text{WSI} > 0.66$ then severe.

So in example $\text{WSI} = 0.25$ hence we consider it mild

But as RI is severe so with mild we can put warning

RI Mild + inflammation warning
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Example 2

$$\text{count} - \text{moderate} = 0.5$$

$$\text{Density} - \text{Mpld} = 0$$

$$\text{Redness} = \text{severe} = 1$$

$$\begin{aligned}\Rightarrow \text{WSI} &= (0.5) \times (0.5) + (0) \times (0.25) + (1) \times (0.25) \\ &= 0.25 + 0 + 0.25 \\ &= 0.5\end{aligned}$$



Moderate

Example-3

$$\text{count} - \text{severe} = 1$$

$$\text{Density} - \text{moderate} = 0.5$$

$$\text{RI} - \text{moderate} = 0.5$$

$$\begin{aligned}\Rightarrow \text{WSI} &= (1) \times (0.5) + (0.5) \times (0.25) + (0.5) \times (0.25) \\ &= 0.5 + 0.125 + 0.125 \\ &= 0.75\end{aligned}$$



severe

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II] Pigmentation.

Clinical Logic for Weights

- ① % Pigmentation Area → most imp. → How much skin is affected. → 0.5
- ② Pigmentation Intensity → shows severity of darkness → second most imp → 0.25
- ③ Melanin Index (MI) → clinical scale → supports validation

Similar to acne → Numeric score for Pigmentation

Mild → 0

Moderate → 0.5

Severe → 1

Asperour Defined Band Definitions

Area → mild < 10%, moderate 10-30%, severe > 50

Δ Gray → mild < 20, moderate 20-25, severe > 50

MI → < 50, moderate 50-100, severe > 100

$$\text{Similarly} \Rightarrow 0.5 + 0.25 + 0.25 = 1 = \frac{1}{3} = 0.33$$

So if ^{Total} severity < 0.33 → Mild

0.33 < 0.66 → Moderate

> 0.66 → Severe.

Example 1

Suppose Area = 8% $\rightarrow$ Mild $\rightarrow 0$

Δ Gray = 55 $\rightarrow$ severe $\rightarrow 1$

MT = 120 $\rightarrow$ severe $\rightarrow 1$

$$\begin{aligned}\text{Then; Total severity} &= (0) \times (0.5) + (1) \times (0.25) + (1) \times (0.25) \\ &= 0 + 0.25 + 0.25 \\ &= 0.5 \rightarrow \underline{\text{moderate}}\end{aligned}$$

Moderate with High MT

Example-2

Suppose Area = 35% $\rightarrow$ severe $\rightarrow 1$

Δ Gray = 40 $\rightarrow$ moderate $\rightarrow 0.5$

MT = 80 $\rightarrow$ moderate $\rightarrow 0.5$

$$\begin{aligned}\text{Then Total severity} &= (1) \times (0.5) + (0.5) \times (0.25) + (0.5) \times (0.25) \\ &= 0.5 + 0.125 + 0.125 \\ &= 0.75\end{aligned}$$

$\downarrow$

severe

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III] wrinkles

Weights

- ① Wrinkle count : Mild < 5 , Moderate $5-10$, severe > 10
(per cm²)
- ② Depth (Δ Gray) Mild < 20 , Moderate $20-50$, severe > 50
- ③ Length (in mm) < 20 mild, $20-40$ moderate, > 40 severe

Numeric scores for wrinkle

mild - 0, moderate - 0.5, severe - 1.0

Severity classification

< 0.33 mild

$0.33 - 0.66$ moderate

> 0.66 severe

Example

count = 3 (mild 0)

Depth = 60 (severe 1)

Length = 15 (mild 0)

$$\begin{aligned}\text{Total severity} &= (0)(0.5) + (1)(0.25) + (0)(0.25) \\ &= 0.25 \\ &= \text{mild} \approx \text{severe Depth.}\end{aligned}$$

$$\textcircled{2} \text{ count} = 2 \quad (m, 12, 0)$$

$$\text{Depth} = 55 \quad (\text{serve}, 1)$$

$$\text{Length} = 45 \quad (\text{serve}, 1)$$

$$\text{Total strength} = \cancel{(0)(0.5)} + \cancel{0.2} \\ (0)(0.5) + (1)(0.25) + 1(0.25)$$

$$= 0.50$$

$$= \text{serve}$$